JAVASCRIPT | ASSIGNMENT 4

Loop Control Structures

Task: Write a for loop to print a multiplication table up to 5. Use a while loop with a break statement to find the first number greater than 10 that is divisible by 6, and a do...while loop that skips numbers using continue.

CODE

```
// Assignment 4

// 1. for loop: multiplication table of 5
console.log("-- Multiplication Table of 5 --");
for (let i = 1; i <= 5; i++) {
  console.log("5 x " + i + " = " + 5 * i);
}

// 2. while loop with break: first number > 10 divisible by 6
console.log("-- First number > 10 divisible by 6 --");
let n = 11;
while (true) {
  if (n % 6 === 0) {
    console.log("Found: " + n);
    break;
  }
  n++;
}

// 3. do...while loop with continue: skip even numbers
console.log("-- Odd numbers 1 to 9 (skipping evens) --");
let j = 0;
do {
  j++;
  if (j % 2 === 0) {
    continue; // skip even numbers
  }
  console.log(j);
} while (j < 9);
```

OUTPUT

```
-- Multiplication Table of 5 --
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
-- First number > 10 divisible by 6 --
Found: 12
-- Odd numbers 1 to 9 (skipping evens) --
1
3
5
7
9
```

NOTES

- break exits the loop completely, while continue skips only the current iteration.
- In a do...while loop, j++ must come before continue, otherwise the loop never ends.
- The expected output in the assignment does not show the do...while part, so skipping even numbers is used as the example.